

Evaluating LLaMA LLMs and Prompt Engineering for Educational Applications

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Abstract

This poster discusses the evaluation of prompt engineering strategies in large language models (LLMs) through an analysis of response accuracy, reproducibility, and latency across multiple model variants. The study included 174 prompt-response trials across ten categories, including logic, mathematics, and language interpretation, administered to three versions of Meta's LLaMA models (1B, 3B, and 8B parameters). Each trial measured response accuracy, reproducibility, and latency (using Python code to quantify the latter two systematically) to assess how model scale influences both output quality and efficiency. Results reveal distinct scaling patterns: the LLaMA 8B model achieved the highest accuracy (75.86%) and reproducibility (0.38) but demonstrated longer latency (24.27 seconds on average), whereas smaller models responded faster but with lower accuracy (24.14% for 1B and 44.83% for 3B). Across all models, the average accuracy was 48.28%, reproducibility 0.35, and mean latency 14.78 seconds. These findings highlight a measurable trade-off between reasoning reliability and speed. The significance of this work extends to computing education, where LLMs are increasingly integrated as teaching and tutoring tools. Understanding how prompt design and model scale affect the consistency of generated responses is critical for developing trustworthy and transparent educational AI-based systems. By analyzing LLM behavior across structured prompts, this study provides insight into how educators and developers can optimize model interactions for classroom learning, automated feedback, and other pedagogical applications.

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1 Further Information

1.1 1. Introduction

Large Language Models such as LLaMA 3 are being rapidly incorporated into educational technologies: from intelligent tutoring systems to lesson and content creation. However, their performance

can vary depending on prompt phrasing and model size. This project investigates how prompt engineering and model scale affect three key dimensions of model behavior: accuracy, reproducibility, and latency. The goal of this project is to inform educators and developers on how to select and structure LLMs for reliable educational applications.

1.2 2. Methodology

Three versions of Meta's LLaMA model - LLaMA 3 (8B), LLaMA 3.2 (3B), and LLaMA 3.2 (1B)- were tested using the open-source Ollama platform. A total of 174 prompts, most of which were from a Kaggle dataset [6], were distributed across ten categories: mathematics, logic, geography, history, fiction, law, common misconceptions, politics/economics, open-ended, superstitions, and science. For each prompt, the following metrics were recorded: **Accuracy** was categorized as correct, incorrect, or reasonable based on predefined answers; this was done manually to ensure that responses were contextually accurate. Correct responses conveyed the same content as the expected answer; incorrect responses were clearly wrong or unrelated; and reasonable responses were partially correct or contextually appropriate but not fully precise. **Reproducibility** was measured as the average lexical similarity between three repeated responses ($n=3$) to the same prompt, using word-level, normalized, case-insensitive Jaccard similarity (0-1). Repeated prompts were submitted sequentially with a 1-second interval between runs. **Latency** was defined as the time (in seconds) from prompt submission to response completion, measured via Python subprocess timing. Data collection and computation of reproducibility and latency were fully automated using Python scripts.

1.3 3. Results

The results obtained from the study highlight model and category-level performance across the evaluated metrics. Accuracy (%) represents the percentage of responses classified as "correct" out of all responses for that model or category. Although responses were originally categorized as correct, reasonable, or incorrect, only the fully "correct" responses were used to calculate this percentage. Metrics in reference to the various models are summarized in Table 1, while metrics corresponding to a subset of six evaluated prompt categories are summarized in Table 2.

2 Discussion and Educational Implications

The results demonstrate clear trends across both model size and task category with important implications for educational applications.



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Table 1: Performance metrics for different LLaMA models.

Model	Accuracy (%)	Reproducibility	Latency (s)
LLaMA 3 (8B)	75.86	0.38	24.27
LLaMA 3.2 (3B)	44.83	0.34	9.85
LLaMA 3.2 (1B)	24.14	0.33	6.23

Table 2: Performance metrics by task category.

Category	Accuracy (%)	Reproducibility	Latency (s)
Math	66.7	0.53	10.6
Politics/Econ	66.7	0.35	13.2
Open-Ended	66.7	0.32	19.6
Science	29.6	0.34	17.7
Law	33.3	0.24	15.6
Geography	25.0	0.32	8.5

Note: Values represent averages for all prompts evaluated across the three LLaMA models.

Larger models, particularly LLaMA 3 (8B), achieved the highest accuracy (75.86% correct responses) and reproducibility (0.38), though with longer average latency (24.27 s). This suggests that larger models are better suited for applications requiring reliable and accurate outputs, such as intelligent tutoring systems, automated grading, or personalized learning support. Smaller models were faster but less reliable, demonstrating the trade-off between responsiveness and quality. Longer latency in larger models may impact real-time interactive learning, such as automated tutoring. Smaller models can provide fast feedback for practice exercises, but with reduced accuracy.

Analysis by category shows that more structured reasoning tasks, such as Math, Politics/Econ, and Open-Ended prompts, had higher accuracy (66.7%) and reproducibility (up to 0.53), whereas knowledge-intensive or interpretive categories, including Science, Law, and Geography, were more challenging (accuracy 25.0–33.3%). This indicates that LLMs may be more effective in supporting subjects with clear step-by-step problem-solving or open-ended reasoning, while factual recall or domain-specific content may require additional scaffolding, such as integrated reference materials or retrieval-augmented systems.

Reproducibility scores highlight that model consistency varies along with the category of the prompt. Prompts with defined structure and expectations (e.g., math) produced more repeatable outputs, which is critical for maintaining fairness and reliability in educational settings. It is important to note that randomness in model generation was not controlled, and repeated prompts were tested sequentially, which may have influenced reproducibility.

It is also important to consider potential risks when deploying LLMs in educational contexts. Incorrect or biased outputs could mislead students, while prompts containing sensitive data may raise privacy concerns. Mitigation strategies include human oversight, validation of model outputs against reliable sources, and integrating these models as a supplement rather than a replacement for educational guidance. By acknowledging these limitations, educators and developers can deploy LLMs responsibly while minimizing potential harms.

From an educational perspective, these findings suggest suitable methods for deploying LLMs in learning environments. Larger models can enhance instructional quality for tasks requiring consistent and accurate guidance, while smaller models may provide faster support. Knowledge-intensive subjects may benefit more from hybrid approaches, such as combining LLM outputs with other content or teacher oversight, to ensure accuracy. Future educational systems could utilize these findings to effectively use varying LLM model sizes based on task type: using larger models for explanation or grading, and smaller ones for quick feedback and lower-stakes practice.

Finally, the study is limited by the relatively small number of prompts per category and the subjectivity in manually classifying responses in accuracy (e.g. distinguishing ‘reasonable’ from ‘correct’). Future research could explore broader prompt sets, automated methods for classification of accuracy, controlled randomness in model generation, more detailed statistical analyses, qualitative evaluation of reasoning errors, and methods to integrate LLMs into curriculum-structured learning tools to maximize their educational impact.

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